

LUC BROGAT-MOTTE

Postdoctoral researcher in machine learning ◊ luc.brogat.motte@gmail.com

EDUCATION

- ENS Paris-Saclay** 2018–2019
M.S. MVA (Mathématiques, Vision, Apprentissage), graduated with honors (16.8/20).
- Télécom Paris** 2016–2019
Engineering degree in applied mathematics (probability, statistics, scientific computing), GPA: 3.98/4.
- CPGE MPSI/MP* — Lycée Marcelin Berthelot** 2013–2016
Intensive undergraduate program in mathematics and physics.

EXPERIENCE

- Postdoctoral Researcher, Istituto Italiano di Tecnologia** 2025–
Supervised by Lorenzo Rosasco (MaLGa, University of Genoa).
Research on the estimation of continuous-time stochastic dynamical systems.
Contributed to teaching the “Sequential Prediction and Reinforcement Learning” course at the University of Genoa.
Co-organizer of the MaLGa Seminar Series (15 seminars).
- Member, COST Action CA24136 (InterCoML)** 2025–
European working group on interactions between control theory and machine learning.
- Postdoctoral Researcher, CentraleSupélec** 2023–2025
Supervised by Riccardo Bonalli (L2S, CentraleSupélec) and Alessandro Rudi (SIERRA, INRIA Paris).
Research on the estimation of controlled stochastic differential equations.
- PhD in Mathematics and Computer Science, Télécom Paris** 2019–2023
Supervised by Florence d’Alché-Buc (Télécom Paris) and Juho Rousu (Aalto University).
Research topics: statistical learning theory, kernel methods, optimal transport.
Runner-up for the Best PhD Thesis Award (CSDAI, Institut Polytechnique de Paris), 2024.
- Reviewer for international journals and conferences** 2019–
JMLR, Constructive Approximation, SIAM Journal on Mathematics of Data Science,
Artificial Intelligence Journal, Systems & Control Letters,
NeurIPS, AISTATS, ECML PKDD.
- Teaching Assistant, Télécom Paris** 2019–2023
Led tutorials and designed practical sessions in statistics and machine learning.
- Research Intern (Master’s), Télécom Paris & Aalto University** Apr.–Sept. 2019
Learning output representations in structured prediction with applications to molecular prediction.
- Digital Technician, National Institute of Integrative Medicine, Melbourne** July 2017
Web development, computer graphics, and hardware installation.

PUBLICATIONS AND PREPRINTS

- Fast learning rates for physics-informed kernel methods.** (Preprint)
Luc Brogat-Motte, Joachim Bona-Pellissier, Giacomo Meanti, Lorenzo Rosasco.
- Learning controlled stochastic differential equations.** (Preprint)
Luc Brogat-Motte, Riccardo Bonalli, Alessandro Rudi.
- Safely learning controlled stochastic dynamics.** (NeurIPS 2025)
Luc Brogat-Motte, Alessandro Rudi, Riccardo Bonalli.

Sketch in, sketch out: accelerating both learning and inference for structured prediction with kernels. (AISTATS 2024)

Tamim El Ahmad, Luc Brogat-Motte, Pierre Laforgue, Florence d'Alché-Buc.

Learning to predict graphs with fused Gromov–Wasserstein barycenters. (ICML 2022)

Luc Brogat-Motte, Rémi Flamary, Céline Brouard, Juho Rousu, Florence d'Alché-Buc.

Vector-valued least-squares regression under output regularity assumptions. (JMLR 2022)

Luc Brogat-Motte, Alessandro Rudi, Céline Brouard, Juho Rousu, Florence d'Alché-Buc.

Duality in RKHSs with infinite-dimensional outputs: application to robust losses. (ICML 2020)

Pierre Laforgue, Alex Lambert, Luc Brogat-Motte, Florence d'Alché-Buc.

TEACHING

Teaching contribution, University of Genoa

2025

10 hours in the M2 course “Sequential Prediction and Reinforcement Learning.”

Teaching assistant, Télécom Paris

2019–2023

130.5 hETD (L3–M2) in statistics and machine learning. Courses include: Statistics (L3), Machine Learning (M1), Kernel Machines (M2), Structured Data: Learning and Prediction (M2), Structured Prediction (Executive MSc).

STUDENT SUPERVISION

PhD student (previously Master’s research intern)

Luca Piccaluga, Forecasting algorithms for continuous-time stochastic dynamical systems.

Co-supervised with Lorenzo Rosasco.

June 2025–

Master’s research intern

Hubert Leroux, *A-PIKS: Active Learning for Physics-Informed Kernel Methods*.

Co-supervised with Joachim Bona-Pellissier and Lorenzo Rosasco.

Apr.–Aug. 2026

Master’s research intern

Florentin Acker, *Physics-Informed Kernel Learning with Misspecified Physical Constraints*.

Co-supervised with Joachim Bona-Pellissier and Lorenzo Rosasco.

Apr.–Aug. 2026

SELECTED PRESENTATIONS

WCCM-ECCOMAS 2026, Munich, Germany

July 2026

Oral presentation, “Learning Controlled Stochastic Differential Equations” (MS237: Recent Advances in Scientific Machine Learning for Differential Equations and Control Problems).

Laboratoire de l’Informatique du Parallélisme (LIP), ENS de Lyon

Apr. 2026

Invited research seminar.

Rencontres de Statistiques du CEREMADE, Université Paris-Dauphine

Dec. 2025

Invited talk.

NeurIPS 2025, San Diego, USA

Dec. 2025

Poster presentation.

Invited research talks

Sept.–Oct. 2025

I2M Marseille (Statistics Seminar); IMT Toulouse (Statistics and Optimization Seminar);

LAAS-CNRS (POP team, Toulouse); GIPSA-Lab (GAIA team, Grenoble);

LIS (QARMA team, Marseille); Inria Rennes (MALT team).

Isaac Newton Institute, Cambridge, UK

July 2025

Invited seminar, “Representing, calibrating & leveraging prediction uncertainty” workshop.

CSDAI Best PhD Thesis Awards*June 2024*

Presentation as recipient of the CSDAI Best PhD Thesis Award (runner-up).

ICML 2022, Baltimore, USA*July 2022*

5-minute oral and poster presentation.

ELLIS Theory Workshop, Arenzano, Italy*July 2022*

Poster presentation.

IDIA-IP Paris Day, École Polytechnique*June 2022*

Poster presentation.

UMR MIA Seminar, AgroParisTech*June 2022*

Research seminar talk.

LIKE22 Workshop*Jan. 2022*

Contributed talk.